Cell Design Specification VBF-T2R1SINGLEMODEL-DS-01

Virtual Battery Factory — t2_r1_singlemodel — 2026-08-25

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> Case: t2_r1_singlemodel (grid energy storage). Every value mechanically taken from the
> parameter set / simulation outputs; sources annotated per line. Generation date 2026-08-25.

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 / graphite (Li-ion), Chen2020 base parameter set	parameter set (entry 0 meta.base_params)
Nominal capacity	5.0 Ah (parameter set) / 6.9555 Ah simulated 1C discharge	parameter set `Nominal cell capacity [A.h]` / cell/r3_1c_dfn.json:capacity_ah
Voltage window	2.5 – 4.2 V	parameter set lower/upper cut-off
Cell dimensions	65 mm x 1580 mm x 0.390 mm (layer stack incl. collectors); shell thickness Not provided	parameter set height/width/layer thicknesses
Electrolyte formulation	EC/EMC + LiPF6 (parameter set default), c_e0 1000 mol/m3; additive pack: FEC, VC, PS, EP, LiDFOB-anion, LiDFP-anion	parameter set / funnel dispositions
Cation transference number	0.2594	parameter set
Midpoint voltage	3.574 V	cell/r3_energy.json:midpoint_voltage_v
DC resistance	0.0124 ohm	cell/r3_energy.json:dcr_ohm

2. Electrode and separator

Layer	Thickness (um)	Porosity	Active-material vol. frac.	Particle radius (um)	Collector
Positive	100	0.335	0.665	3.0	Al 16 um
Separator	12.0	0.47	—	—	—
Negative	250	0.25	0.75	3.5	Cu 12 um

N/P ratio = 1.48 (no capacity-density keys in Chen2020; full theoretical lithiation range (c_max*F/3600 cancel), c_max neg 33133 / pos 63104 mol/m3)

3. Process design parameters

Parameter	Value	Formula	Notes
Positive areal density	216.9 g/m2	thickness x (1-porosity) x density	parameter set
Negative areal density	310.7 g/m2	thickness x (1-porosity) x density	parameter set
Positive compaction density	2.17 g/cm3	density x (1-porosity) / 1000	parameter set
Negative compaction density	1.24 g/cm3	density x (1-porosity) / 1000	parameter set
Electrolyte fill amount	0.0 g	pore volume x 1.2 g/cm3 x fill 1.0	literature electrolyte density 1.2 g/cm3 (annotated)
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	—	design recommended value; production-line value requires tuning

4. Mass breakdown (calc-energy contract caliber, electrolyte excluded)

Layer	kg/m2	kg
Positive active layer	0.0000	0.0 g
Negative active layer	0.0000	0.0 g
Current collectors	0.0000	0.0 g
Separator	0.0025	0.3 g
Total	—	**69.9 g**

Electrolyte excluded from mass (parameter set lacks electrolyte density) — cell/r3_energy.json:electrolyte_included=false.

5. Performance verification

Item	Value	Criterion (entry 0)	Verdict
Energy density	358.01 Wh/kg	>= 327.18	PASS
SEI thickness @100 cyc	194.56 nm	<= 500 nm	PASS
SEI thickness @500 cyc	439.74 nm	<= 550 nm	PASS
-20 C retention	99.59 %	>= 90 %	PASS
4C fast charge plating	ap_min +0.112 V (45 C start); +0.111 V (25 C conservative)	plated == false	PASS
4C max temperature	342.5 K	monitored only (no red line in entry 0)	INFO
5C discharge	0.4939 Ah (7% of 1C; voltage collapses to cut-off in ~71 s — severely rate-limited; informational, no 5C criterion)	informational (no 5C criterion)	INFO

6. Design notes

Parameters changed vs Chen2020 baseline and why (reasoning traceable to log evaluate/plan entries):
1. Negative electrode thickness 250 um / positive 100 um (baseline thin electrodes): areal flux reduction for 4C reaction-zone crowding; DFN adopted after SPMe c_e-profile artifact was diagnosed (plan-update entry).
2. Negative electrode exchange-current density = 2.0 A/m2 (baseline value overridden): D-scan optimum for 4C plating suppression (non-monotonic; i0=20 A/m2 fails). Bridge: interfacial kinetics from the additive pack.
3. Positive/negative particle radii 3.0/3.5 um: surface-area tuning supporting the kinetics fix.
4. SEI pack: k_sei 3e-13 m/s, SEI i0 7.5e-8 A/m2, EC diffusivity 1e-19 m2/s (baseline D_ec 2e-18): growth is diffusion-limited at L>100 nm (Yang2017 ec reaction limited), L ~ sqrt(D_ec); the D_ec value is a film-property extension of the SEI-suppression bridge, recorded as estimate

(plan-update entries document the 1e-18 -> 2e-19 -> 8e-20 -> 1e-19 correction chain; 8e-20 was dropped because its 500-cycle aging dies deterministically at cycle ~174, IDA_CONV_FAIL).